Institute of Engineering
Pulchowk Campus
Department of Electronics and Computer Engineering
Master of Science in Information & Communication Engineering

Course Title : **Research & Optimization Methods**
Course Code :
Credit : 4
Class Load : 4 hours
Evaluation :

Course Objectives:

This course objective is to introduce the research methodology concepts and also optimization techniques. It will provide the basic understanding of various concepts and popular methods used in quantitative techniques, operations research and research methodology.

Course Contents :

1. **Quantitative Techniques**
   Statistical model building, Sampling, correlation, inference, causation, prediction, Estimation, confidence intervals, estimating regression parameters, multiple regression analysis, selection of predictor variables, prediction and confidence intervals, model building, qualitative variables, stepwise regression, residual analysis, Hypothesis testing Forecasting using Time series models, Choosing the appropriate forecasting technique, moving average, exponential smoothing, etc., ANOVA Analysis.

2. **Operations Research**
   Linear Programming Model formation, Graphical solution, Simplex method, sensitivity analysis and duality, Transporation and assignment problems, Network Optimization problems, Integer Programming

3. **Research Methodology**
   Problem identification, Literature Review, Research Design, Data & Experimental Environment setup, Hypothesis formulation and testing, Data Analysis, Result Interpretation & Visualization, Report preparation and Presentation, Referencing styles.

Reference Book:

NO particular single book will cover the contents. Any books on the above contents could be referenced, however, for quick reference here is the list:

1. Introduction to Operations Research : Frederick S. Hillier, Gerard J. Liberman : latest Edition
2. Statistics for Engineering and the Sciences : William Mendenhall, Terry Sincich, Latest Edition
3. Social Science Research and Thesis Writing : Prem R. Pant, Latest Edition